Discrete differential geometry in homotopy type theory

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Summary

Summary

This work brings to HoTT

- connections, curvature, and vector fields
- the index of a vector field
- a theorem in dimension 2 that total curvature = total index

Classical → HoTT

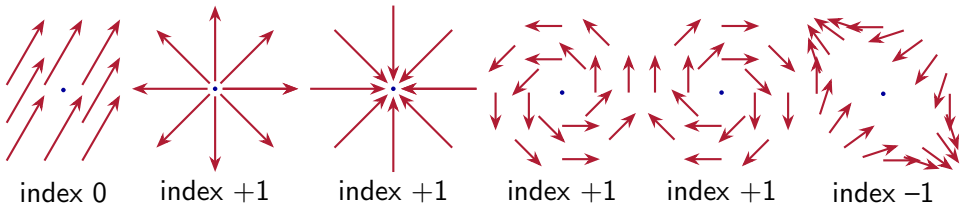
Let M be a smooth, oriented 2-manifold without boundary, F_A the curvature of a connection A on the tangent bundle, and X a vector field with isolated zeroes $x_1, \ldots, x_n$.

$$\begin{array}{ccccc} \frac{1}{2\pi}\int_M F_A & = & \sum_{i=1}^{n} \mathrm{index}_X(x_i) & = & \chi(M) \\ \downarrow & & \downarrow & & \\ \sum_{\text{faces } F} \flat_F & = & \sum_{\text{faces } F} L_F^X & & \end{array}$$

Classical index

Near an isolated zero there are only three possibilities: index 0, 1, –1.

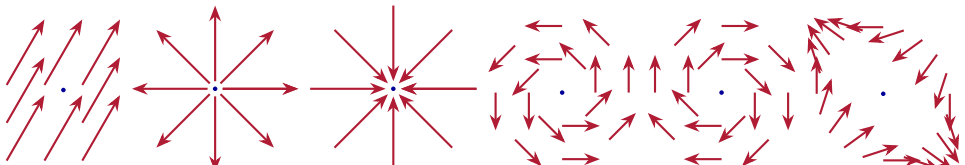
Index is the winding number of the field as you move clockwise around the zero.



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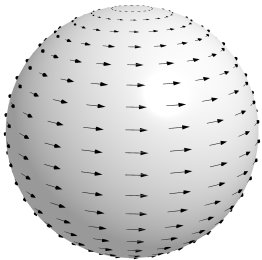


(This definition is using an implicit ad-hoc connection near the zero, more on this later.)

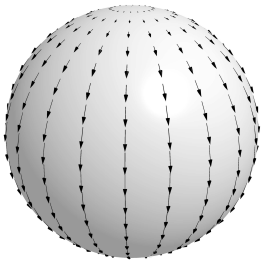
Poincaré-Hopf theorem

The total index of a vector field is the Euler characteristic.

Examples:



Rotation: index $+1$ at each pole $= \mathbf{2}$



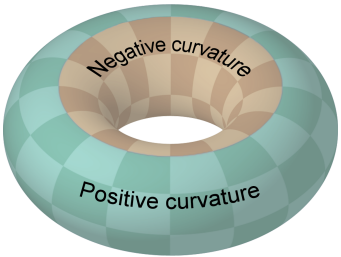
Height: index $+1$ at each pole $= \mathbf{2}$

Gauss-Bonnet theorem

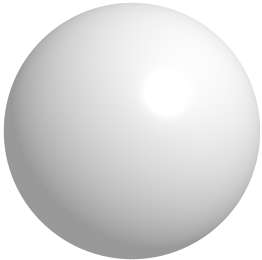
Total curvature divided by 2π is the Euler characteristic.

Curvature in 2D is a function $F_A : M \to \mathbb{R}$.

$\int_M F_A$ sums the values at every point.



Positive and negative curvature cancel: **0**



Constant curvature 1, area 4π: **2**

Plan

- Combinatorial manifolds
- Torsors and classifying maps
- Connections and curvature
- Vector fields
- Main theorem

HoTT background

1. Symmetry,
   Bezem, M., Buchholtz, U., Cagne, P., Dundas, B. I., and Grayson, D. R., (2021-)
   `https://github.com/UniMath/SymmetryBook`.
2. Central H-spaces and banded types,
   Buchholtz, U., Christensen, J. D. , Flaten, J. G. T., and Rijke, E. (2023)
   arXiv:2301.02636
3. Nilpotent types and fracture squares in homotopy type theory,
   Scoccola, L. (2020)
   MSCS 30(5). arXiv:1903.03245

Combinatorial manifolds

Manifolds in HoTT

- Recall the classical theory of simplicial complexes
- Define a realization procedure to construct types

Simplicial complexes

Definition

An abstract simplicial complex M of dimension n is an ordered list of sets $M \stackrel{\text{def}}{=} [M_0, \ldots, M_n]$ consisting of

- a set M_0 of vertices
- sets M_k of subsets of M_0 of cardinality $k+1$
- downward closed: if $F \in M_k$ and $G \subseteq F$, $|G| = j+1$ then $G \in M_j$

We call the truncated list $M_{\leq k} \stackrel{\text{def}}{=} [M_0, \ldots, M_k]$ the k-skeleton of M.

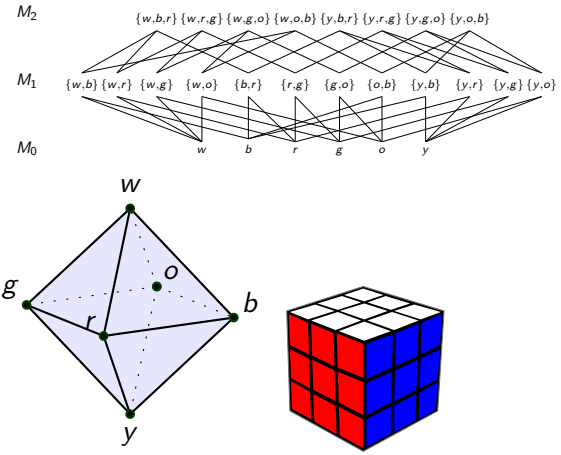
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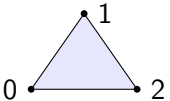
Example

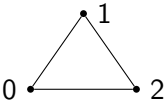
The complete simplex of dimension n, denoted $\Delta(n)$, is the set $\{0, \ldots, n\}$ and its power set. The $(n-1)$-skeleton $\Delta(n)_{\leq(n-1)}$ is denoted $\partial\Delta(n)$ and will serve as a combinatorial $(n-1)$-sphere.

$\Delta(1)$ is visually

, $\partial\Delta(1)$ is visually

,

$\Delta(2)$ is visually 

, $\partial\Delta(2)$ is visually 

Homotopy realization: dimension 0

We will realize simplicial complexes by means of a sequence of pushouts.

It is likely we are exactly applying the $\int$ modality, but we won't need that.

Base case: the realization $\mathbb{M}$ of a 0-dimensional complex M is M_0.

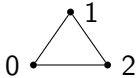
In particular the 0-sphere $\partial\Delta(1) \stackrel{\text{def}}{=} \partial\Delta(1)_0$.

Homotopy realization: dimension 1

For a 1-dim complex $M \overset{\text{def}}{=} [M_0, M_1]$ the realization is given by

$$\begin{array}{ccc} M_1 \times \partial\Delta(1) & \xrightarrow{\mathrm{pr}_1} & M_1 \\ {\scriptstyle \mathbb{A}_0}\downarrow & \overset{h_1}{\nearrow} & \downarrow{\scriptstyle *_{\mathbb{M}_1}} \\ M_0 = \mathbb{M}_0 & \longrightarrow & \mathbb{M}_1 \end{array}$$

Homotopy realization: dimension 1

For example the simplicial 1-sphere $\partial\Delta(2) \overset{\text{def}}{=}$ 

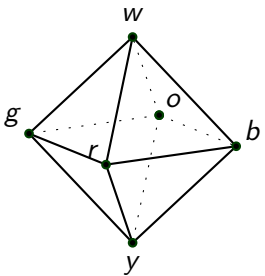
is given by

$$\begin{array}{ccc} \partial\Delta(2)_1 \times \partial\Delta(1) & \longrightarrow & \partial\Delta(2)_1 \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \partial\Delta(2)_0 & \longrightarrow & \partial\Delta(2) \end{array} \quad \text{i.e.} \quad \begin{array}{ccc} \{\{0,1\},\{1,2\},\{2,0\}\} \times \{0,1\} & \longrightarrow & \{\{0,1\},\{1,2\},\{2,0\}\} \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \{0,1,2\} & \longrightarrow & \partial\Delta(2) \end{array}$$

Homotopy realization: dimension 1

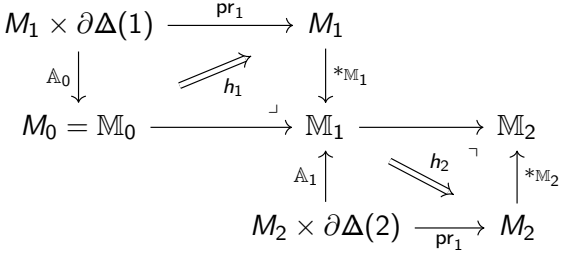
Or the 1-skeleton of the octahedron $\mathbb{O}$:

$$\begin{array}{ccc} \{\{w,g\},\ldots\}\times\{0,1\} & \longrightarrow & \{\{w,g\},\ldots\} \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \{w,g,\ldots\} & \longrightarrow & \mathbb{O}_1 \end{array}$$



Homotopy realization: dimension 2

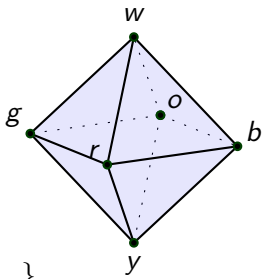
To realize $M \overset{\text{def}}{=} [M_0, M_1, M_2]$ use $\partial\Delta(1), \partial\Delta(2)$:



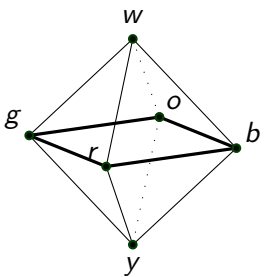
Homotopy realization: dimension 2

The full octahedron $\mathbb{O}$:

$$\begin{array}{ccccc}
\{\{w,g\},\ldots\}\times\{0,1\} & \xrightarrow{\mathrm{pr}_1} & \{\{w,g\},\ldots\} & & \\
\downarrow & \overset{h_1}{\Rightarrow} & \downarrow & & \\
\{w,g,\ldots\} & \longrightarrow & \mathbb{O}_1 & \longrightarrow & \mathbb{O}_2 \\
 & & \uparrow & \overset{h_2}{\Rightarrow} & \uparrow \\
 & & \{\{w,g,r\},\ldots\}\times\partial\Delta(2) & \xrightarrow[\mathrm{pr}_1]{} & \{\{w,g,r\},\ldots\}
\end{array}$$



Homotopy realization: dimension 2



$\leftarrow \text{link}(w)$

The link of a vertex w in a 2-complex is: the sets not containing w but whose union with w is a face.

A combinatorial manifold is a simplicial complex all of whose links are* simplicial spheres.

This will be our model of the tangent space.

*the (classical) geometric realization is homeomorphic to a sphere

Combinatorial manifolds $\leftrightarrow$ smooth manifolds

Theorem (Whitehead (1940))

Every smooth n-manifold has a compatible structure of a combinatorial manifold: a simplicial complex of dimension n such that the link is a combinatorial $(n-1)$-sphere, i.e. its geometric realization is an $(n-1)$-sphere.

`https://ncatlab.org/nlab/show/triangulation+theorem`

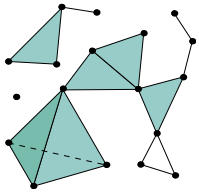
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Counterexample: Wikipedia says this is a simplicial complex, but we can see it fails the link condition:



Torsors

What type families $\mathbb{M} \to \mathcal{U}$ will we consider? Families of torsors, also called principal bundles.

Torsors

Let G be a (higher) group.

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- Let BG be the type of G-torsors.
- Let G_{reg} be the G-torsor consisting of G acting on itself on the right.

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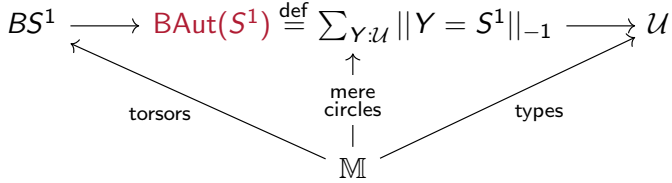
1. $\Omega(BG, G_{\text{reg}}) \simeq G$ and composition of loops corresponds to multiplication in G.
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3. 1 & 2 $\implies$ BG is a $\mathsf{K}(G, 1)$.

See the Buchholtz et al. H-spaces paper for more.

How to map into BS^1

To construct maps into BS^1 we lift a family of mere circles.

families of:

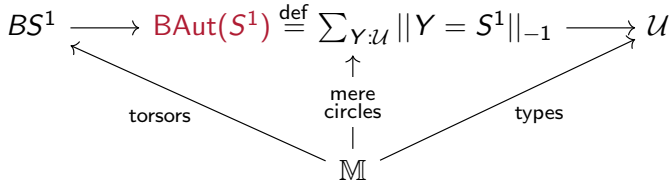


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Other names:

- $\mathrm{BAut}(S^1) = BO(2) = \mathrm{EM}(\mathbb{Z}, 1)$ (where $\mathrm{EM}(G, n) \stackrel{\text{def}}{=} \mathrm{BAut}(\mathrm{K}(G, n))$)
- $BS^1 = BSO(2) = \mathrm{K}(\mathbb{Z}, 2)$

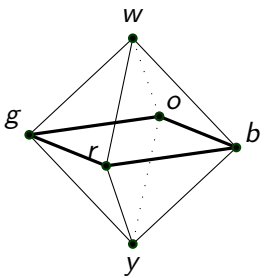
Connections and curvature

Connections

Connections are extensions of a bundle to higher skeleta.

They live in a single dimension.

Recall link



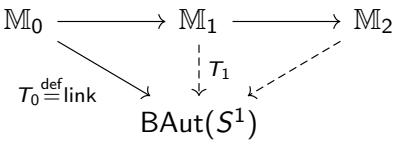
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Define the tangent bundle on a combinatorial manifold to be $T_0 \stackrel{\text{def}}{=} \text{link} : \mathbb{M}_0 \to \text{BAut}(S^1)$.

Connections on the tangent bundle

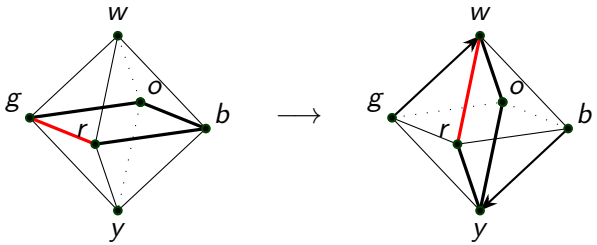
An extension T_1 of T_0 to $\mathbb{M}_1$ is called a connection on the tangent bundle.



$T_1 : \mathbb{M}_1 \to \mathrm{BAut}(S^1)$ extending link

We will define T_1 on the edge wb, so we need a term $T_1(wb) : \mathrm{link}(w) =_{\mathrm{BAut}(S^1)} \mathrm{link}(b)$.

We imagine tipping:

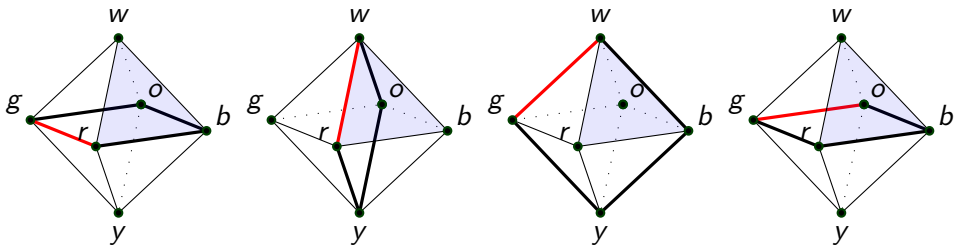


$T_1(g : \mathrm{link}(w)) \stackrel{\mathrm{def}}{=} w : \mathrm{link}(b), \ldots$.

Use this method to define T_1 on every edge.

$T_1 : \mathbb{M}_1 \to \mathrm{BAut}(S^1)$ extending link

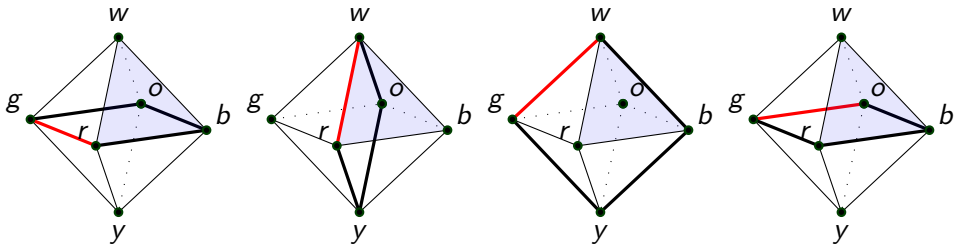
Denote the path $wb \cdot br \cdot rw$ by $\partial(wbr)$. Consider $T_1(\partial(wbr))$:



We come back rotated by 1/4 turn. Call this rotation $R : \mathrm{link}(w) =_{BAut(S^1)} \mathrm{link}(w)$.

Extending T_1 to a face

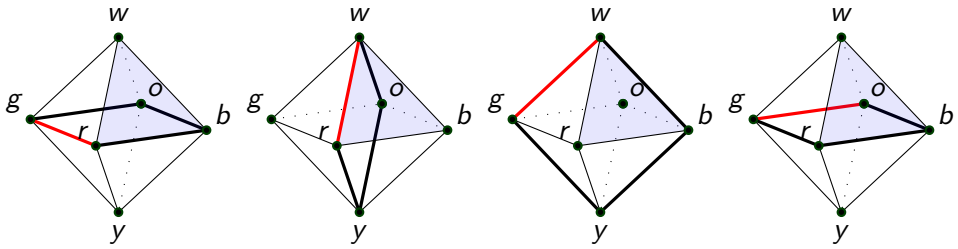
Let $H_{wbr} : \mathrm{refl}_w =_{w=_{\mathbb{M}} w} \partial(wbr)$ be the filler homotopy of the face.



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T_2 must live in $T_1(\mathrm{refl}_w) =_{(\mathrm{link}(w) =_{\mathrm{BAut}(S^1)} \mathrm{link}(w))} T_1(\partial(wbr)) = R$

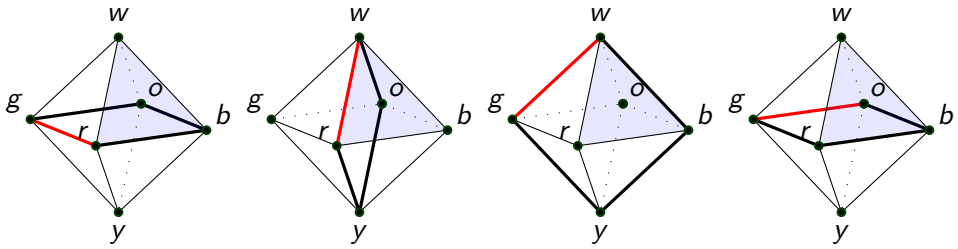


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T_2 must be a homotopy $H_R : \mathrm{id} = R$ between automorphisms of $\mathrm{link}(w)$.



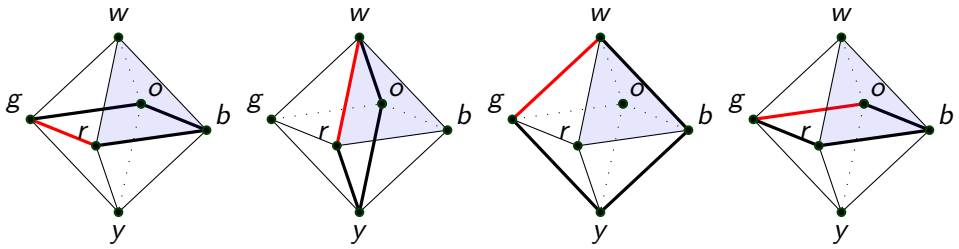
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For example, a path $H_R(g) : g = Rg = o$. Choose go.



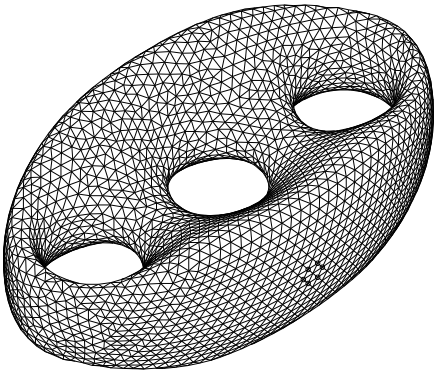
Not everything is a regular solid

The octahedron is very symmetrical. The links are all squares and the tippings rotate by 90 degrees.

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A more typical situation has variable links, averaging to 6 sides. Also transport will rotate by small amounts.

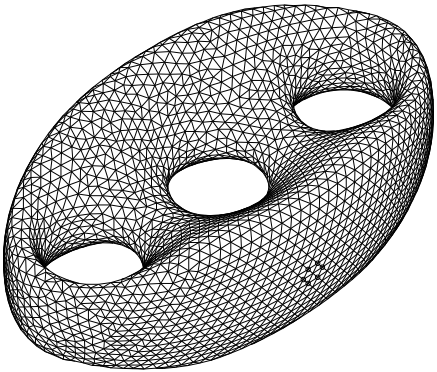


Not everything is a regular solid

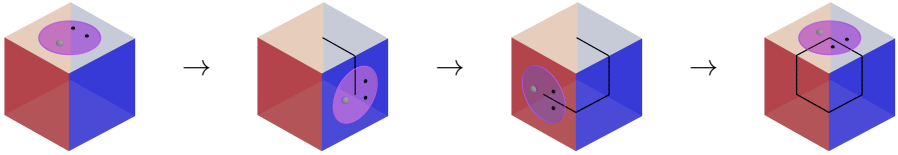
The octahedron is very symmetrical. The links are all squares and the tippings rotate by 90 degrees.

A more typical situation has variable links, averaging to 6 sides. Also transport will rotate by small amounts.

Idea: introduce a subdivider D_n which takes a k-gon to an nk-gon, adding n vertices along each edge. Set $T_0(v) \stackrel{\text{def}}{=} D_{n(v)} \circ \text{link}(v)$. Make all tangent spaces the same LCM-sided polygon. Also transport should now be able to imitate small angles.



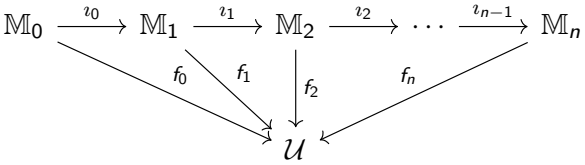
Original inspiration



The definition of a connection

Definition

If $\mathbb{M} \stackrel{\text{def}}{=} \mathbb{M}_0 \xrightarrow{\imath_0} \cdots \xrightarrow{\imath_{n-1}} \mathbb{M}_n$ is the realization of a combinatorial manifold and all the triangles commute in the diagram:

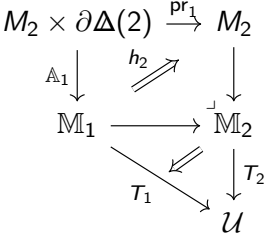


- The map f_k is a k-bundle on $\mathbb{M}$.
- The pair given by the map f_k and the proof $f_k \circ \imath_{k-1} = f_{k-1}$, i.e. that f_k extends f_{k-1} is called a k-connection on the $(k-1)$-bundle f_{k-1}.

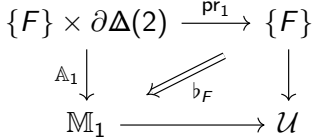
The definition of curvature

Definition (cont.)

An extension consists of M_2-many extensions to faces:



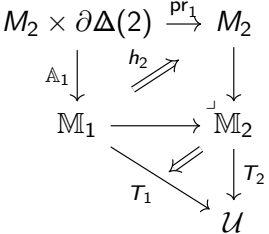
Here's the outer square for a single face F:



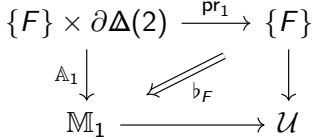
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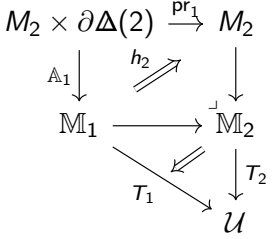


$T_1(\partial(F))$ is the curvature at the face F and the filler $\flat_F : \mathrm{id} = T_1(\partial F)$ is called a flatness structure for the face F.

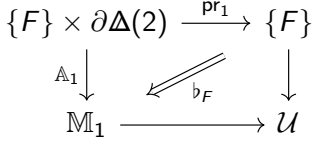
The definition of curvature

Definition (cont.)

An extension consists of M_2-many extensions to faces:



Here's the outer square for a single face F:



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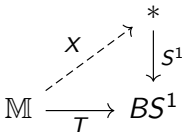
The distinction between the path $\flat_F$ and the endpoint $T_1(\partial(F))$ is small enough to be confusing.

Vector fields

Vector fields

Let $T : \mathbb{M} \to BS^1$ be an oriented tangent bundle on a 2-dim realization of a combinatorial manifold.

- Our bundles of mere circles can only model nonzero tangent vectors.
- A global section of this family would be a trivialization of T, so that's not a good definition.

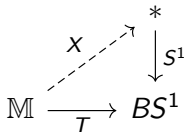


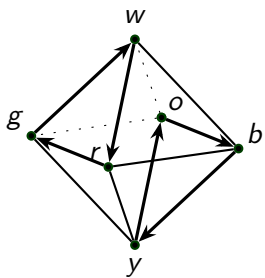
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Our solution:





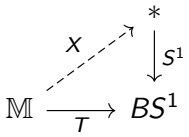
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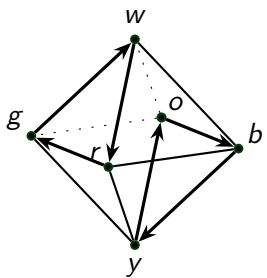
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- A vector field is a term $X : \prod_{m:\mathbb{M}_1} Tm$.





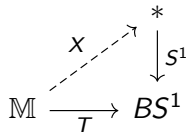
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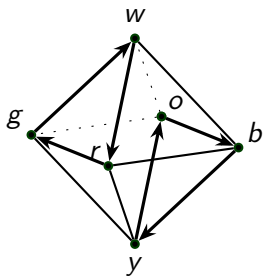
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- A vector field is a term $X : \prod_{m:\mathbb{M}_1} Tm$.
- It models a classical nonvanishing vector field on the 1-skeleton.

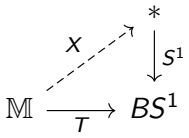




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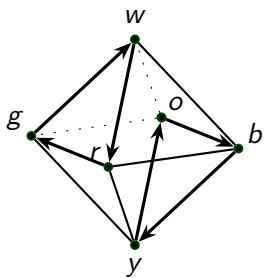
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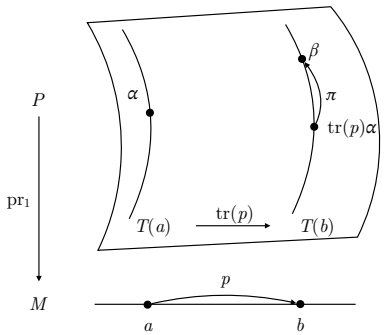


Our solution:

- A vector field is a term $X : \prod_{m:\mathbb{M}_1} Tm$.
- It models a classical nonvanishing vector field on the 1-skeleton.
- We model classical zeros by omitting the faces.

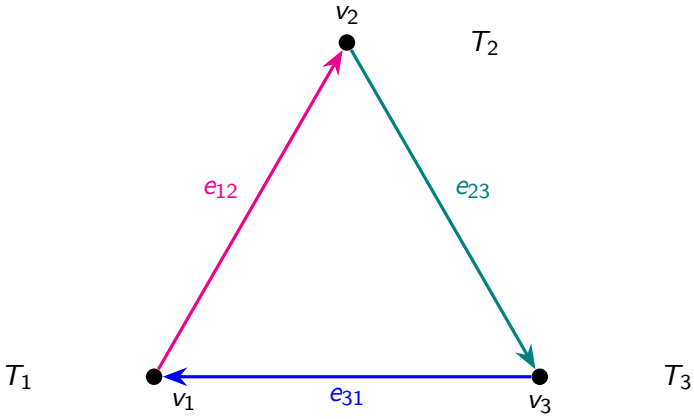


Reminder: pathovers

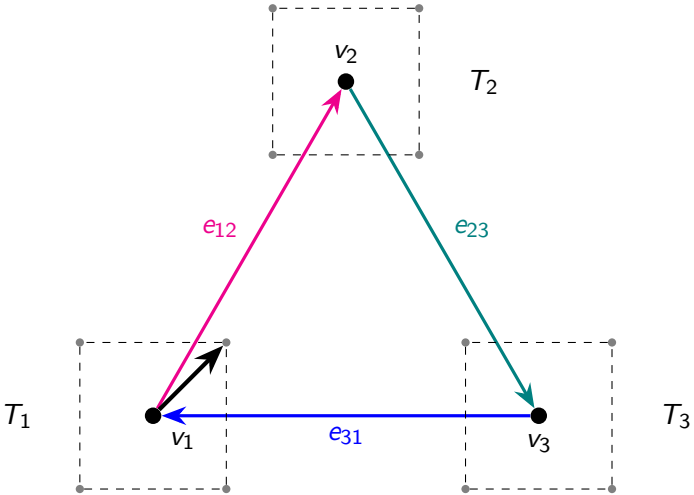


- Recall pathovers (dependent paths).
- There is an asymmetry: we pick a fiber to display π, the path over p.
- Dependent functions map paths to pathovers: $\mathsf{apd}(X)(p) : \mathsf{tr}_p(X(a)) = X(b)$ (simply denoted $X(p)$).

Next goal: define the index of a vector field on a face by computing $X(\partial F)$ around a face.

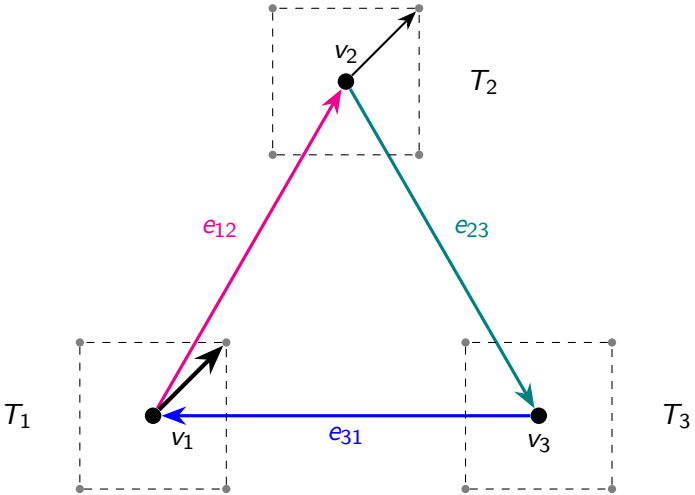


An example of swirling and index at this face.



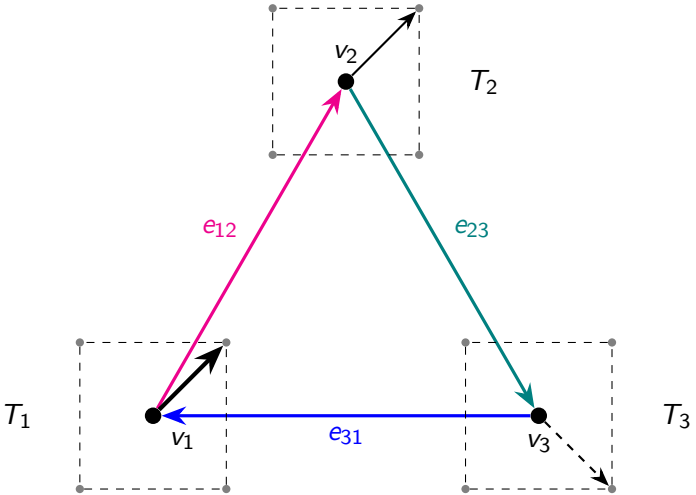
An example of swirling and index at this face.

- Denote by X_1 this vector $X(v_1) : T_1$.
-
-
-



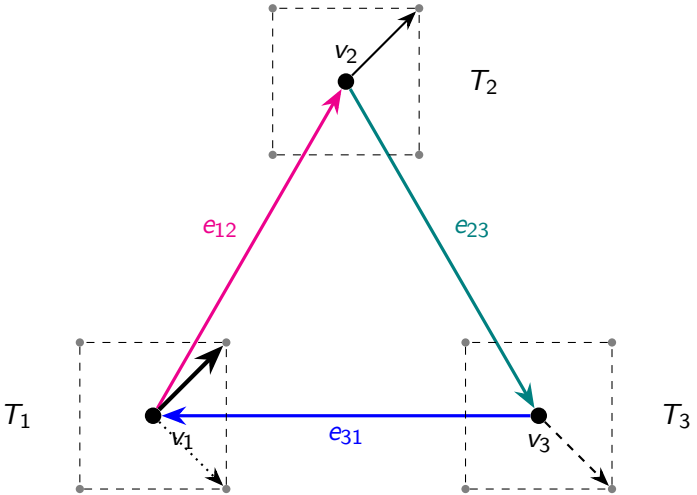
An example of swirling and index at this face.

- Denote by X_1 this vector $X(v_1) : T_1$.
- Say T_{21} is trivial. Denote the transported vector as thinner.
-
-



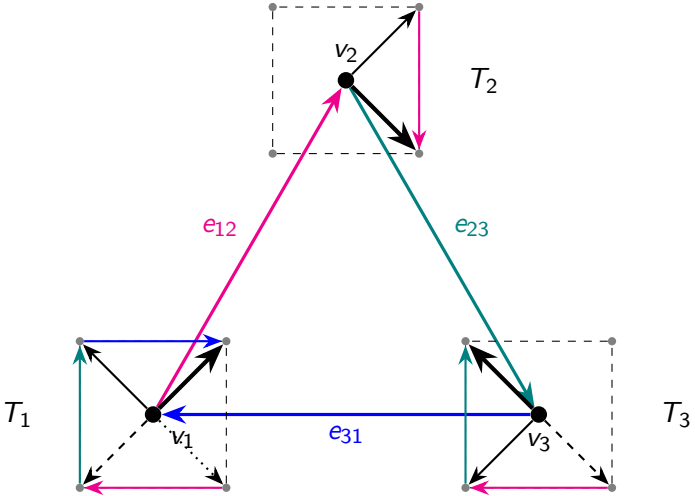
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- Denote by X_1 this vector $X(v_1) : T_1$.
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-

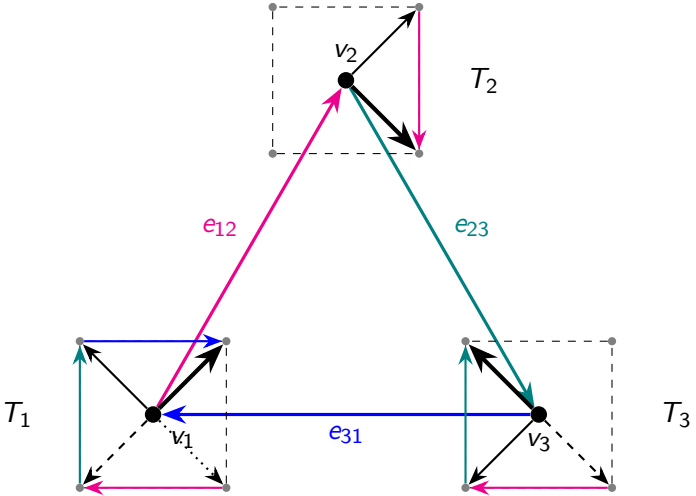


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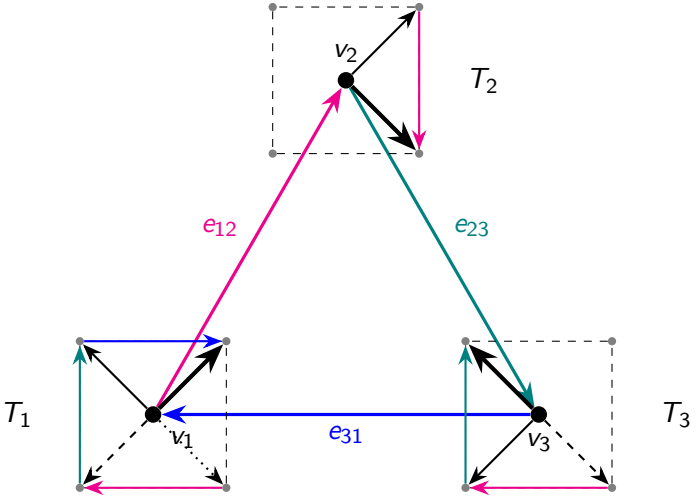
- Denote by X_1 this vector $X(v_1) : T_1$.
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- Say T_{13} is trivial. The thrice-transported vecor is dotted.



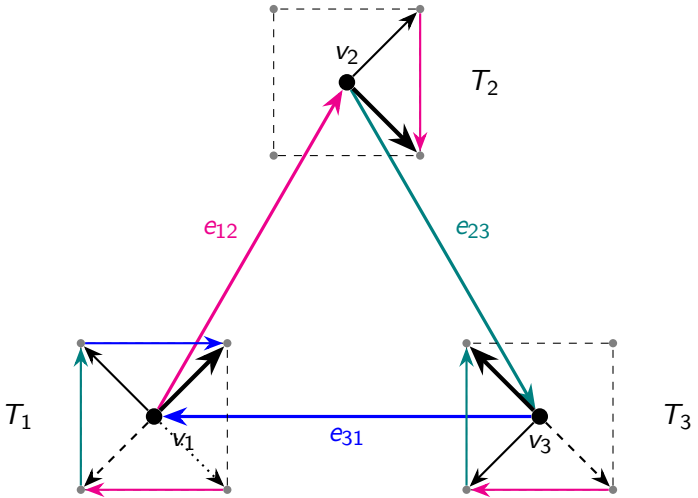
- X on e_{12} is red, etc.



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- We translated all pathover data to the end of the loop.

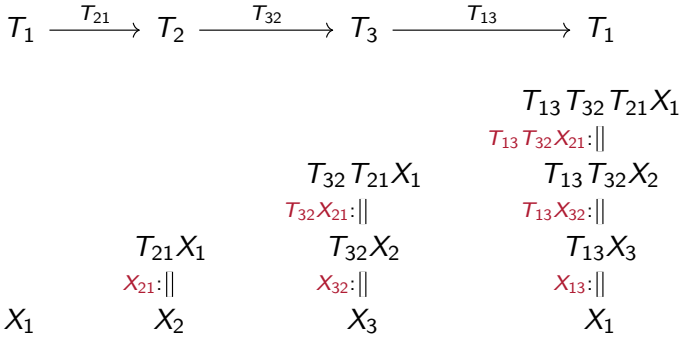


- X on e_{12} is red, etc.
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- (Reminds me of scooping ice cream towards the last fiber.)



- X on e_{12} is red, etc.
- We translated all pathover data to the end of the loop.
- (Reminds me of scooping ice cream towards the last fiber.)
- The total pathover $X(\partial F)$ is called the swirling X_F of X at the face F.

Symbolic version



Index

$$\mathrm{tr}_F \stackrel{\mathrm{def}}{=} \mathrm{tr}(\partial F) \quad : T_1 =_{BS^1} T_1 \qquad \text{curvature}$$

$$\flat_F \stackrel{\mathrm{def}}{=} \flat(\partial F) \quad : \mathrm{id} =_{(T_1 =_{BS^1} T_1)} \mathrm{tr}_F \qquad \text{flatness}$$

$$X_F \stackrel{\mathrm{def}}{=} X(\partial F) \quad : \mathrm{tr}_F(X_1) =_{T_1} X_1 \qquad \text{swirling}$$

Index

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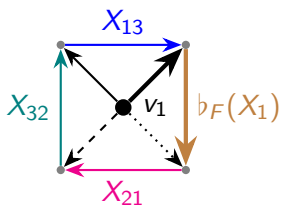
$$X_F \stackrel{\text{def}}{=} X(\partial F) \quad : \mathrm{tr}_F(X_1) =_{T_1} X_1 \qquad \text{swirling}$$

Definition

The flattened swirling of the vector field X on the face F is the loop

$$L_F^X \stackrel{\text{def}}{=} \flat_F(X_1) \cdot X_F : (X_1 =_{T_1} X_1).$$

The index of the vector field X on the face F is the integer I_F^X such that $\mathrm{loop}^{I_F^X} =_{S^1} \alpha(L_F^X)$ for some iso $\alpha : T_1 \simeq S^1$.



Index

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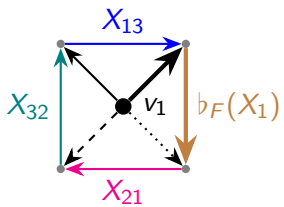
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The classical index uses a trivial local connection in a chart. We use the given type family and then subtract the effect of curvature.



Main theorem

Simplifying swirling

Swirling involves concatenating dependent paths. Can we simplify that?

Pay off all our assumptions 1: torsor structure, vector field

$$T_1$$

$$T_{13}\,T_{32}\,T_{21}X_1$$

$$T_{13}\,T_{32}X_{21}:\Vert$$

$$T_{13}\,T_{32}X_2$$

$$T_{13}X_{32}:\Vert$$

$$T_{13}X_3$$

$$X_{13}:\Vert$$

$$X_1$$

Pay off all our assumptions 1: torsor structure, vector field

T_1

- Def: $\alpha_i \overset{\text{def}}{=} s(-, X_i) : T_i \xrightarrow{\sim} S^1$ (trivialization on 0-skeleton).

$T_{13}\, T_{32}\, T_{21} X_1$

$T_{13}\, T_{32} X_{21}:$ ‖

$T_{13}\, T_{32} X_2$

$T_{13} X_{32}:$ ‖

$T_{13} X_3$

$X_{13}:$ ‖

X_1

Pay off all our assumptions 1: torsor structure, vector field

T_1

$T_{13}\,T_{32}\,T_{21}X_1$

$T_{13}\,T_{32}\,X_{21}$:

$T_{13}\,T_{32}X_2$

$T_{13}X_{32}$:

$T_{13}X_3$

X_{13}:

X_1

- Def: $\alpha_i \overset{\text{def}}{=} s(-, X_i) : T_i \xrightarrow{\sim} S^1$ (trivialization on 0-skeleton).
- Def: $\rho_{ji} \overset{\text{def}}{=} \alpha_j(T_{ji}(X_i))$ is the rotation of T_{ji}.

$$\begin{array}{ccc} T_i & \xrightarrow{T_{ji}} & T_j \\ \text{base}\mapsto X_i \;\circlearrowright\; \downarrow \alpha_i & & \downarrow \alpha_j \;\circlearrowright\; \text{base}\mapsto X_j \\ S^1 & \xrightarrow[(-)+\rho_{ji}]{} & S^1 \end{array}$$

Pay off all our assumptions 1: torsor structure, vector field

$$
\begin{array}{c}
T_1 \\
\\
T_{13}\,T_{32}\,T_{21}\,X_1 \\
{\scriptstyle T_{13}\,T_{32}\,X_{21}:}\,\| \\
T_{13}\,T_{32}\,X_2 \\
{\scriptstyle T_{13}\,X_{32}:}\,\| \\
T_{13}\,X_3 \\
{\scriptstyle X_{13}:}\,\| \\
X_1
\end{array}
$$

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$$
\begin{array}{ccc}
T_i & \xrightarrow{T_{ji}} & T_j \\
{\scriptstyle \text{base} \mapsto X_i}\ \big(\ {\scriptstyle \alpha_i}\ \big\downarrow & & \big\downarrow\ {\scriptstyle \alpha_j}\ \big)\ {\scriptstyle \text{base} \mapsto X_j} \\
S^1 & \xrightarrow[(-)+\rho_{ji}]{} & S^1
\end{array}
$$

- Lemma: $\rho_{ij} = -\rho_{ji}$ because in T_j:
  $\rho_{ij} + \rho_{ji} + X_j = \rho_{ij} + T_{ji}X_i = T_{ji}(\rho_{ij} + X_i) = T_{ji}\,T_{ij}X_j = X_j$.

Pay off all our assumptions 1: torsor structure, vector field (cont.)

$$T_1$$

$$T_{13}\,T_{32}\,T_{21}X_1$$

$$T_{13}\,T_{32}X_{21}:\ \|$$

$$T_{13}\,T_{32}X_2$$

$$T_{13}X_{32}:\ \|$$

$$T_{13}X_3$$

$$X_{13}:\ \|$$

$$X_1$$

Pay off all our assumptions 1: torsor structure, vector field (cont.)

T_1

- Define $\sigma_{ji} \stackrel{\text{def}}{=} \alpha_j(X_{ji}) : \rho_{ji} =_{S^1}$ base,.

$T_{13}\,T_{32}\,T_{21}X_1$

$T_{13}\,T_{32}X_{21}$: ‖

$T_{13}\,T_{32}X_2$

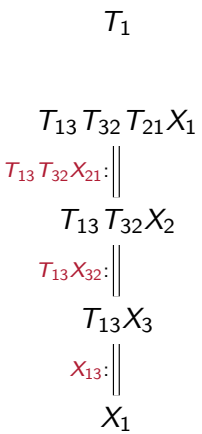
$T_{13}X_{32}$: ‖

$T_{13}X_3$

X_{13}: ‖

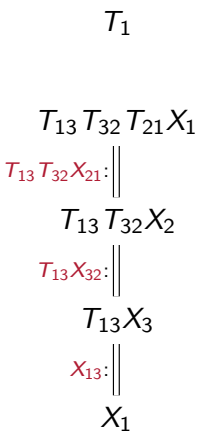
X_1

Pay off all our assumptions 1: torsor structure, vector field (cont.)



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- Paths of the form $(a =_{S^1} \text{base})$ can be added:
  - $+ : (a = \text{base}) \times (b = \textit{base}) \to (a + b = \textit{base})$.
  - $p + q = (p + b) \cdot q$.

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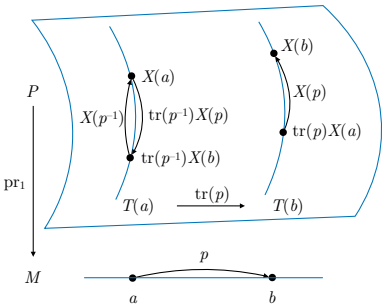
$$
\begin{array}{c}
T_1 \\
\\
T_{13}\,T_{32}\,T_{21}X_1 \\
T_{13}\,T_{32}\,T_{21}X_{21}\colon \| \\
T_{13}\,T_{32}X_2 \\
T_{13}X_{32}\colon \| \\
T_{13}X_3 \\
X_{13}\colon \| \\
X_1
\end{array}
$$

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  $\implies \sigma_{ij} + \sigma_{ji} = \text{refl}_{\text{base}}$ (T_{ij} just translates X_{ji} to cat with X_{ij}).

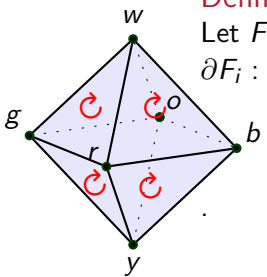
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Pay off all our assumptions 2: no boundary, commutativity

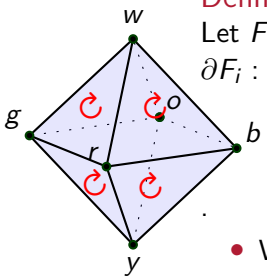


Definition

Let $F_1, \ldots, F_n$ be the faces of $\mathbb{M}$, $v_i : F_i$ be designated vertices, and $\partial F_i : v_i = v_i$ be the triangular boundaries. The total swirling is

$$X_{\text{tot}} \stackrel{\text{def}}{=} \sigma_{\partial F_1} + \cdots + \sigma_{\partial F_n}$$

Pay off all our assumptions 2: no boundary, commutativity



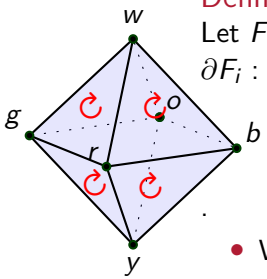
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- We assume that this expression involves every edge once in each direction.
- S^1 is commutative, hence complete cancellation.

Consequence

$$\mathrm{tr}_F \overset{\text{def}}{=} \mathrm{tr}(\partial F) \quad : T_1 =_{BS^1} T_1 \quad \text{curvature}$$

$$\flat_F \overset{\text{def}}{=} \flat(\partial F) \quad : \mathrm{id} =_{(T_1 =_{BS^1} T_1)} \mathrm{tr}_F \quad \text{flatness}$$

$$X_F \overset{\text{def}}{=} X(\partial F) \quad : \mathrm{tr}_F(X_1) =_{T_1} X_1 \quad \text{swirling}$$

$$L_F^X \overset{\text{def}}{=} \flat_F(X_1) \cdot X_F \quad : (X_1 =_{T_1} X_1) \quad \text{flattened swirling}$$

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These can all be totaled in S^1 to give

$$\mathrm{tr}_{\mathrm{tot}} \overset{\mathrm{def}}{=} \sum_i \rho_{\partial F} = \mathrm{base} \qquad X_{\mathrm{tot}} \overset{\mathrm{def}}{=} \sum_i \sigma_{\partial F} = \mathrm{refl}_{\mathrm{base}}$$

$$\flat_{\mathrm{tot}} \overset{\mathrm{def}}{=} \sum_i \flat_{\partial F} \qquad L_{\mathrm{tot}}^X \overset{\mathrm{def}}{=} \sum_i \flat_{\partial F} + \sigma_{\partial F} = \sum_i \flat_{\partial F}$$

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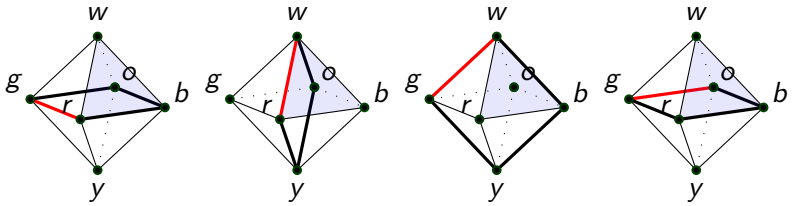
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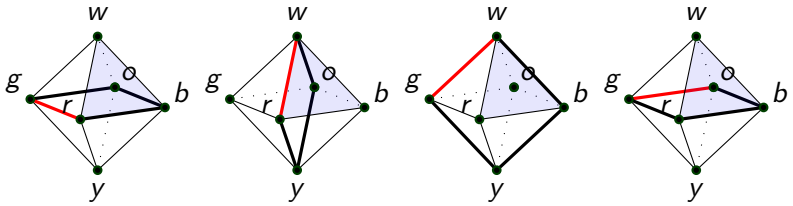
So in our lingo: the total flatness equals the total flattened swirling. □

Examples

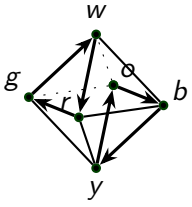


Each face contributes $\flat_F = H_R$, a 1/4-rotation. Total: 2.

Examples

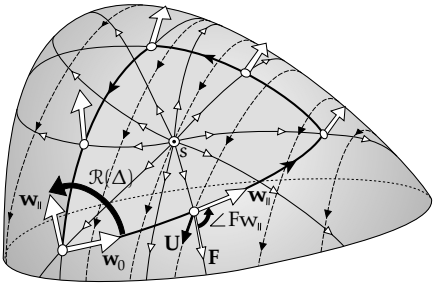


Each face contributes $\flat_F = H_R$, a 1/4-rotation. Total: 2.



For total index one obtains $+1$ from F_{wrg}, $+1$ from F_{ybo}, $+0$ from others. Total: 2.

Classical proof



[26.2] *The difference* $\mathcal{R}(\Delta) - 2\pi\mathcal{I}_F(s)$ *can be found by summing over the edges* K_j *the change* $\Phi(K_j)$ *in the illustrated angle* $\angle \mathrm{Fw}_{||}$*, i.e., the rotation of* $\mathbf{w}_{||}$ *relative to* $\mathbf{F}$.

Figure: Needham, T. (2021) Visual Differential Geometry and Forms.

- The classical proof is discrete-flavored.
- "$\angle Fw_{||}$" looked a lot like a pathover.
- Hopf's Φ is defined on edges, not loops. We imitated that too.

Thank you!

Appendix: Conjectural dictionary

Conjectionary

Homotopy realization likely amounts to the shape operator.

> *[G]iven a "cell complex" presentation of a classical topological space, if we can convert it into both a specification for a HIT and a colimit decomposition of that space that issufficiently "cofibrant", then ∫ will preserve that colimit and take the space to the HIT.*
>
> — Mike Shulman, *Brouwer's Fixed Point Theorem in Real-Cohesive HoTT*

Conjectionary

Homotopy realization (and/or $\int$) may provide the following relationships.

Classical octahedron	Homotopy realization, e.g. $\mathbb{O}$
Classical combinatorial manifold M	$\int M$
Derivative	ap
Leibniz rule for $f, g : M \to \mathbb{R}$	Given H-space $(A, *)$, $f, g : X \to A$, $p : x =_X y$ then $\mathrm{ap}(f * g)(p) = \mathrm{ap}(f)(p) * (ga) \cdot (fb) * \mathrm{ap}(g)(p)$. Because in $A \times A$, $(fp, gp) = (fp, \mathrm{refl}) \cdot (\mathrm{refl}, gp)$
The sphere is not flat, as a pointwise statement	Nontrivial flatness on each face

Conjectionary

Connections being "affine", and not (quite) 1-forms	$T_{ji} : T_i = T_j$ being a torsor and not (quite) a group
Space of connections for a given P is contractible.	Two extensions to $\mathbb{O}_1$...
Tensor	?

Conjectionary

Maurer-Cartan form.	Hmm, consider the trivial connection on $M \times G$ or $G \to *$.
Gauge transformations acting on connections and maybe functions (YM) of connections.	
The based gauge group acts freely on connections.	

Conjectionary

Characteristic classes.	$BS^1 \to B^n\mathbb{Z}$.
Chern-Weil theory.	$\mathbb{O} \xrightarrow{T} BS^1 \to B^n\mathbb{Z}$.
Hopf fibration.	$\mathbb{O} \xrightarrow{?} \mathrm{EM}(\mathbb{Z}, 1)$.
Zeros of X = Poincare dual of the Euler class.	